BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester II Examination, 2022–23

Subject: MATHEMATICS
Paper No. MTB-201: Calculus–II

Time Allowed: 3 Hours **Full Marks:** 70

Instructions: *Solve any five questions. Question 1 is compulsory. The figures in the right-hand margin indicate the marks allotted to the respective parts.*

1. **Answer all parts of the following (Compulsory):**

(a) Give ϵ–δ definitions of the limit and continuity of a function $f : \mathbb{R}^2 \to \mathbb{R}$ at a point $(a, b) \in \mathbb{R}^2$. **[2]**

(b) Define $f(x, y) = \frac{x^3y^3}{x^2+y^2}$ for $(x, y) \neq (0, 0)$. Find $\lim_{(x,y)\to(0,0)} f(x, y)$ at the point $(0, 0)$ by using the analytical ϵ–δ definition. **[2]**

(c) Find the equation of the tangent plane to the surface $z = x^2 + y^2 + e^{xy}$ at the point $(1, 0, 2)$. **[2]**

(d) Give the mathematical definition of a homogeneous function $f(x, y, z)$ of degree n. Provide an explicit example of a function $f(x, y, z)$ of fractional degree $3/2$. **[2]**

(e) Let D be a non-empty open set in $\mathbb{R}^2$ and $f : D \to \mathbb{R}$ a real-valued function. Give the definition of the gradient of f at a point (a, b). Find the gradient of the function defined by $f(x, y) = x^2y - xy^2$ for all $(x, y) \in (-2, 2) \times (-2, 2)$ at the coordinate point $(-1, -1)$. **[2]**

(f) State the general multi-variable Mean Value Theorem (MVT) for a function of two independent variables. **[2]**

(g) Show analytically that the infinite integral satisfies the identity:

$$\frac{1}{\sqrt{\pi}} \int_0^\infty e^{-x^2}\, dx = 1$$

[2]

(h) If $u^2 + v^3 = x^2 + y^2$ and $u^2 + v^2 = x^3 + y^3$, then prove that the Jacobian of u and v with respect to independent variables x and y is given by:

$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{1}{2} \cdot \frac{y^2 - x^2}{uv(u - v)}$$

[2]

(i) Evaluate the double integral $\iint_R e^{x^2}\, dx\, dy$, where the bounded integration domain R is defined by:

$$R = \left\{(x, y) \in \mathbb{R}^2 : 2y \leq x \leq 2,\ 0 \leq y \leq 1\right\}$$

[3]

(j) State Fubini's theorem over a rectangular region in $\mathbb{R}^2$. Provide a descriptive example in support of Fubini's theorem. **[3]**

2. (a) Let D be a non-empty set in $\mathbb{R}^2$ and $c = (a, b) \in \mathbb{R}^2$ such that $B_r(c) \setminus \{c\} \subseteq D$ for some radius $r > 0$. Let $f, g : D \to \mathbb{R}$ be functions such that $\lim_{(x,y)\to(a,b)} f(x, y) = \alpha$ and $\lim_{(x,y)\to(a,b)} g(x, y) = \beta$. For any real scalar constant $\gamma \in \mathbb{R}$, show rigorously that:

$$\lim_{(x,y)\to(a,b)} (f + \gamma g)(x, y) = \alpha + \gamma\beta$$

[6]

(b) Define the function $f : \mathbb{R}^2 \to \mathbb{R}$ by:

$$f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$$

Then: (i) discuss the continuity and differentiability of f at the origin $(0,0)$, and (ii) prove that the mixed partials are unequal: $\frac{\partial^2 f(0,0)}{\partial x \partial y} = 1$ and $\frac{\partial^2 f(0,0)}{\partial y \partial x} = -1$. **[6]**

3. (a) Let D be a non-empty open set in $\mathbb{R}^2$ and $f : D \to \mathbb{R}$ a function such that f is differentiable at $(a,b) \in D$. Prove that f must be continuous at (a,b). **[5]**

   (b) State and prove Young's Theorem concerning the symmetry conditions for mixed second-order partial derivatives of a function of two variables. **[7]**

4. (a) Let $xy^3 - x^3y = 6$ be the implicit equation of a planar curve. Find the numerical slope and the explicit structural equation of the tangent line at the point $(1,2)$. **[5]**

   (b) (i) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function of independent variables x and y, and let $x = \phi(u,v)$ and $y = \psi(u,v)$ be functions of independent variables u and v. State the formal definitions of a Jacobian matrix and a Jacobian determinant of f. **[2]**

   (ii) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function of x and y, where $x = r\cos\theta$ and $y = r\sin\theta$. Show that:

$$\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 = \left(\frac{\partial f}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial f}{\partial \theta}\right)^2$$

[5]

5. (a) (i) State Taylor's theorem for the expansion of a multi-variable function $f(x,y)$ of two variables about an expansion center point (a,b). **[2]**

   (ii) Expand the function $f(x,y) = 2x^3 + 3y^3 - 4x^2y$ by Taylor's theorem of maximum order in powers of $(x-1)$ and $(y-2)$. **[5]**

   (b) Find the shortest distance between the linear trajectory line $y = 10 - 2x$ and the standard centered ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$. **[5]**

6. (a) Find the exact total area which is located inside the cardioid boundary $r = 2(1+\cos\theta)$ and outside the bounding circular arc $r = 2$. **[6]**

   (b) State and prove Legendre's Duplication Formula for the classical Gamma function. **[6]**

7. (a) Find the total geometric volume of the standard solid ellipsoid defined by $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. **[6]**

   (b) Let S be a solid closed simplex region (tetrahedron) in the first octant, bounded by the coordinate planes and the plane $x + y + z = 1$. Prove that the Dirichlet triple integral evaluates to:

$$\iiint_S x^{p-1}y^{q-1}z^{r-1}\,dx\,dy\,dz = \frac{\Gamma(p)\Gamma(q)\Gamma(r)}{\Gamma(p+q+r+1)} \quad \text{for } p,q,r > 0$$

[6]

External Portals & Academic Resources:

Examination Portal: https://bhu-exam.vercel.app/ — Student Resource Center: https://quashan-me.vercel.app/
Academic Portfolio: https://me-aryan.vercel.app